

## Information about Individual Contributions

MUST be submitted along with the exam submission

According to Section 15 of the Examination Order, an individual grade or other assessment can only be given in a written group exam if the contribution of each student can be identified (individualisation). If the contribution of each group member is not specified, the exam submission will be rejected.

To assess a group assignment, it is therefore necessary to indicate who is responsible for which sections of your exam submission. This should be specified per section/chapter, not per line. Introduction, partial conclusions, conclusion, and appendices with calculations can be written collaboratively.

Number of characters in each part:

|         |                |
|---------|----------------|
| Part 1: | Approx. 12.000 |
| Part 2: | Approx. 12.000 |
| Part 3: | Approx. 11.000 |

Please indicate the number of students in the group:

3

Please provide information on section responsibility in exam number order:  
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| Exam number: | Responsible for the sections/chapters:                                    |
|--------------|---------------------------------------------------------------------------|
| 72           | 1.1, 1.2.1, 1.2.2, 1.2.4, 1.4.1, 2.3.1, 2.4.2, 5.2.1, 5.2.2, 5.3.3, 5.2.6 |
| 282          | 2.1, 2.5, 1.4.3, 1.4.4, 1.3, 2.3.3, 2.4.3, 5.1.1, 5.2.4                   |
| 16           | 1.5, 2.2, 1.2.3, 1.4.2, 2.4.1, 2.3.2, 5.1.2, 5.1.3, 5.1.4, 5.2.5          |
|              |                                                                           |

## **Declaration of the Use of Generative AI Tools – Economics Programme at the University of Copenhagen**

☒ **I/we have used generative AI as a tool/aid** (*tick the box*)

☐ **I/we have NOT used generative AI as a tool/aid** (*tick the box*)

*If the use of generative AI is permitted in the exam, but you have not used it in your assignment, you only need to tick the box indicating that you have not used GAI, and you do not need to complete the rest of the form.*

**List the generative AI tools used, including a link to the platform (if possible):**

ChatGPT <https://chatgpt.com/>

Perplexity <https://www.perplexity.ai/>

**Briefly describe how generative AI has been used in the assignment (max. 100 words)**

GAI has been used to find relevant literature and to correct grammar and spelling errors. ChatGPT has been used as LaTeX-support e.g. to find errors in code or to tell how to insert a footnote in a table.

We have used GAI for sparring, but the submission solely reflects our own ideas and work.

*NB: GAI-generated content used as a source in the assignment requires correct use of quotation marks and source referencing. Please refer to the guidelines from the University of Copenhagen Library on KUnet [here](#)*

# Part I: Assignment #2

## Relationship Between Inflation and the Interest Rate Spread

## 1.1 Introduction

This paper investigates the dynamic relationship between the interest rate spread and the inflation rate. The causal contemporaneous relationship of these variables are not theoretically clear. Therefore we utilize a Vector Autoregressive Model framework which does not require us to make assumptions on this causal structure. Using a General To Specific approach and maximum likelihood theory we conduct inference to conduct model selection. This approach along with misspecification tests suggest a reduced VAR(12) model as a reasonable model for the data. This estimated model is stable and well specified. Applying this model, we conduct a joint robust Wald-test we conclude that the data does not indicate Granger causality in either direction.

## 1.2 Econometric theory

### 1.2.1 The Vector Auto-Regressive (VAR) Model

We formally define the VAR(k) model by the following stochastic process

$$Z_t = \delta + \Pi_1 Z_{t-1} + \Pi_2 Z_{t-2} + \cdots + \Pi_k Z_{t-k} + \varepsilon_t, \text{ where } \varepsilon_t \sim \text{i.i.d}(0, \Sigma_\varepsilon) \quad (1.1)$$

with  $Z_0, Z_{-1}, \dots, Z_{-(k-1)}$  given.  $Z_t$  is a  $p \times 1$  vector with  $p$  variables,  $\Pi_i$  is a  $p \times p$  matrix with coefficients on  $Z_{t-i}$  (for  $i = 1, 2, \dots, k$ ) and  $\varepsilon_t$  is the  $p \times 1$  vector with stochastic innovations; the error term.

**Stability** Let  $W_t = (Z_t, Z_{t-1}, \dots, Z_{t-k})'$  and the companion matrix  $C$  be given by

$$C \equiv \begin{bmatrix} \Pi_1 & \Pi_2 & \cdots & \Pi_{k-1} & \Pi_k \\ I_p & 0 & \cdots & 0 & 0 \\ 0 & I_p & \cdots & 0 & 0 \\ 0 & 0 & \ddots & 0 & 0 \\ 0 & 0 & \cdots & I_p & 0 \end{bmatrix} \quad (1.2)$$

then we can write (1.1) on the *companion form*:  $W_t = \tilde{\delta} + CW_{t-1} + \tilde{\varepsilon}_t$  and by the well-known result for the VAR(1) model: The VAR(k) model is stable if the companion matrix  $C$  is diagonalizable with all of its eigenvalues inside the complex unit circle.<sup>1</sup>

This also implies that the time series processes contained in  $Z_t$  are weakly stationary - given that the model correctly specifies the actual data generating process.

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<sup>1</sup>Theorem 6.2, Nielsen (2025)

### 1.2.2 (Quasi) Maximum Likelihood Estimation

**MLE.** This paper estimates the parameters using the Gaussian maximum likelihood estimator defined as

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} \log L_T(\theta) \quad (1.3)$$

where  $\Theta$  is the parameter space and  $\log L_T(\theta)$  is the log likelihood function. Formally given by

$$\log L_T(\theta) = \sum_{t=1}^T \ell_t(\theta) = -\frac{Tp}{2} \log(2\pi) - \frac{T}{2} \log(\det(\Sigma)) - \frac{1}{2} \sum_{t=1}^T \epsilon_t(\theta)' \Sigma^{-1} \epsilon_t(\theta) \quad (1.4)$$

, where  $\ell_t(\theta)$  is the log likelihood contribution,  $\epsilon_t(\theta) = Z_t - \mathbb{E}[Z_t \mid \mathcal{I}_{t-1}]$  and  $\Sigma$  is the variance matrix of  $Z_t \in \mathbb{R}^p$ . which is also estimated.

It can be shown that if the likelihood function is correctly specified and certain *high level assumptions* hold the MLE is consistent and asymptotically normal distributed such that<sup>2</sup>

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega_\theta(\theta_0)), \quad T \rightarrow \infty \quad (1.5)$$

**Quasi MLE.** If the likelihood function is just approximately correctly specified such that the score satisfies  $\mathbb{E}[s_t(\theta_0)] = 0$ , where  $s_t(\theta_0) = \frac{\partial}{\partial \theta} \ell_t(\theta)|_{\theta=\theta_0}$  and if the above mentioned assumptions hold, then the estimator is still asymptotically normal and consistent, but with  $\Omega_\theta(\theta_0)$  in (1.5) replaced with the robust variance given by

$$\Omega_\theta^*(\theta_0) = \mathcal{I}(\theta_0)^{-1} \mathcal{J}(\theta_0) \mathcal{I}(\theta_0)^{-1} \quad (1.6^3)$$

In conclusion the QMLE still allows us to conduct inference for hypothesis testing.<sup>4</sup> Note however, that this estimator is not efficient.

### 1.2.3 Misspecification tests

To choose the model with the best properties, we test for misspecification using the test listed in table 2. First of all, it is most important that the model has no autocorrelation, as this would imply an inconsistent estimator.

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<sup>2</sup>Nielsen 2025, theorem 3.1

<sup>3</sup>Where  $\mathcal{I}$  is the fisher-information  $\mathcal{I}(\theta_0) = -\mathbb{E} \left[ \frac{\partial^2 \log \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right]$  and  $\mathcal{J}(\theta_0) = \mathbb{E} \left[ \frac{\partial \log \ell_t(\theta_0)}{\partial \theta} \frac{\partial \log \ell_t(\theta_0)}{\partial \theta'} \right]$

<sup>4</sup>Nielsen 2025, p. 67

<sup>5</sup>Section 2.3 "Formulation and Misspecification Testing", Nielsen (2025)

| Test                                       | Null hypothesis                  | Asymptotic distribution<br>(under $\mathcal{H}_0$ ) <sup>a</sup> |
|--------------------------------------------|----------------------------------|------------------------------------------------------------------|
| Portmanteau test                           | No autocorrelation               | $\chi^2$                                                         |
| No-Heteroskedasticity<br>incl. Cross terms | Unconditional homoscedasticity   | $\chi^2$                                                         |
| Normality test                             | Normally distributed error terms | $\chi^2$                                                         |
| RESET test                                 | Correct functional form          | $\chi^2$                                                         |

<sup>a</sup> For more information on degrees of freedom, see Section 2.3 "Formulation and Misspecification Testing," Nielsen (2025).

**Table 1: Overview of misspecification tests and their null hypotheses<sup>5</sup>**

### 1.2.4 Granger Causality

Formally, a variable  $x$  is said to Granger cause  $y$  if:

$$\mathbb{E}[\eta_h^2|\mathcal{I}_t] < \mathbb{E}[\tilde{\eta}_h^2|\mathcal{I}_t] \quad (1.7)$$

holds for some forecast horizon  $h$ , where  $\eta_h$  and  $\tilde{\eta}_h$  is the forecasting error obtained by forecasting  $y_{t+h}$  using the information set  $\mathcal{I}_t = \{y_t, x_t, y_{t-1}, x_{t-1}, \dots\}$  and  $\tilde{\mathcal{I}}_t = \{y_t, y_{t-1}, \dots\}$  respectively. If this is the case,  $x$  is said to *Granger cause*  $y$ . This does not refer to causality in a structural or deep naturalistic sense. It limits itself to determining whether  $x$  is useful for forecasting  $y$ . Note that this implies a temporal condition where the cause comes before the effect.

**Test granger causality.** Suppose we want to test if  $z_i \in Z$  Granger causes  $z_j \in Z$  ( $i \neq j$ ), where  $Z$  denotes the vector containing all of the endogenous variables. We formalize the null hypothesis by defining the parameter vector  $P_{ij} = (\Pi_{ij}^1, \Pi_{ij}^2, \dots, \Pi_{ij}^k)'$  and test

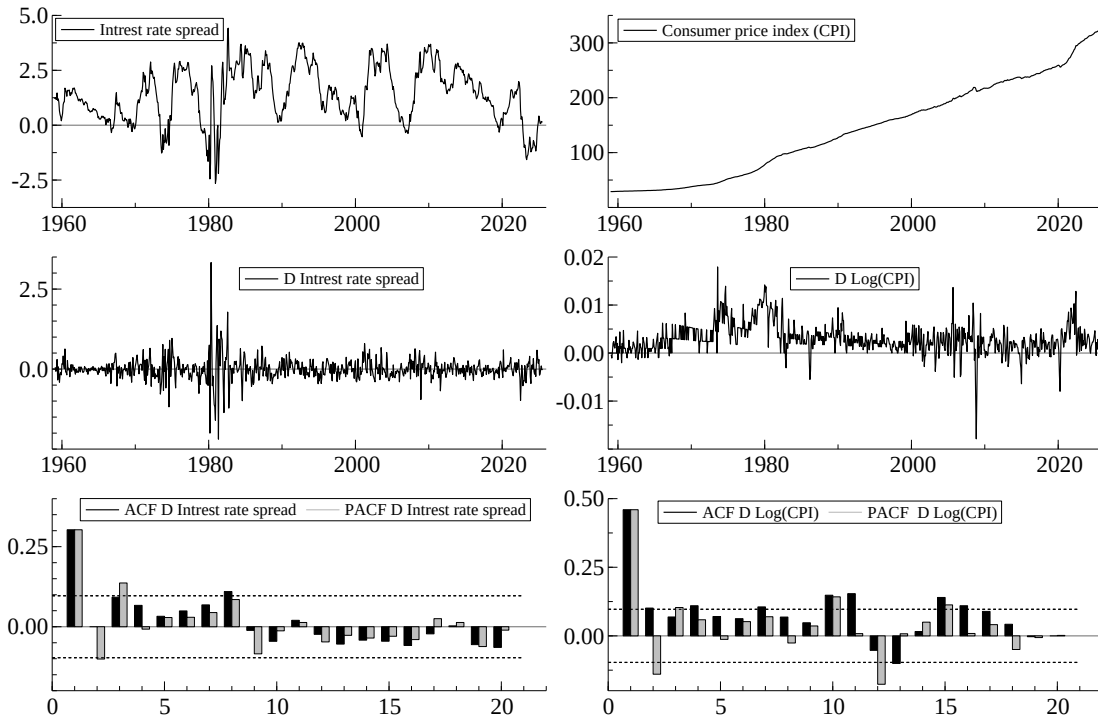
$$\mathcal{H}_0 : P_{ij} = 0 \quad vs. \quad \mathcal{H}_A : P_{ij} \neq 0 \quad (1.8)$$

This jointly tests the statistical significance of lagged variable  $z_j$  on predicting  $z_i$ . Formally we conduct a Wald test using a robust estimation of the covariance matrix (since all of the models considered indicate heteroscedastic error terms). Under the null, the Wald test is asymptotically  $\chi^2(l)$  distributed with  $l$  degrees of freedom, where  $l$  is the number of restrictions.

## 1.3 Presentation of the data

In this paper we consider monthly data on the US Consumer Price Index (CPI) and the Interest rate spread. The data is retrieved from FRED and covers a period from 1959(1) to 2025(7).

A visual examination of the two graphs (See Figure 1) reveals that the interest rate spread is cyclical and highly persistent. The characteristics of the CPI is different, with a clear and upwards turning trend. The characteristics of these level series are clearly not in line with weakly stationary processes.



**Figure 1: Time series properties**

To address the issue of non-stationarity we transform the series to remove trends and unit-root behaviour by taking logs on the CPI and first differences on both series. By graphical inspection of 1 the two series no longer exhibit problematic violations of stationarity. Note, however, that there seems to be some volatility-clustering and outliers. These problems are mostly present in the series before 1990. The rest of the paper will only consider the period from 1990 and onwards. For the above reasons the transformed variables are preferred when modelling the time series.

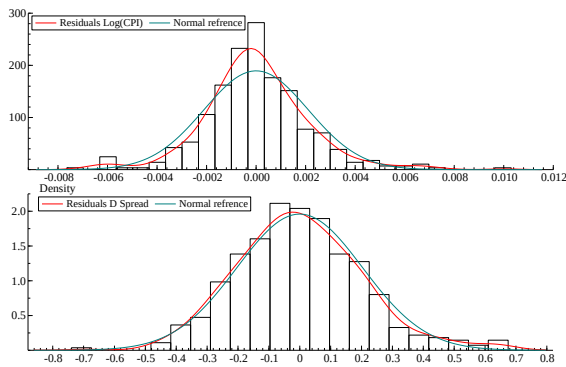
To guide our statistical analysis we compute and plot the (partial) autocorrelation functions (ACF and PACF) from 1990 and onwards (See figure 1). By considering the PACF for the two variables, we find a high level of persistency in the time series. Especially for the inflation rate which show significant partial autocorrelation to about lag 12. However, it is important to note that the PACF is less useful when considering a bivariate VAR model than it is when considering an ARMA model. This is due to the correlations between the two time series. Finally, the empirical tests<sup>1.4</sup> indicate that the model should allow for around 10-12 lags as a general enough starting point which plausibly nests the true model.

## 1.4 Empirical results

### 1.4.1 Model selection

This paper will conduct model selection used the General To Specific (GETS) approach. As mentioned earlier this is motivated by the result that inference is valid if the true DGP is nested as a specific parametrisation of the statistical model, which allows us to test hypothesis on the parameters and reduce the model.

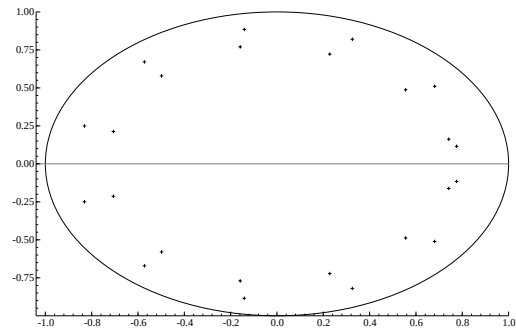
To verify the model specification, we test for autocorrelation, normality, and heteroscedasticity cf. 2.3. Our starting point is the VAR(12) model. This model does not suffer from autocorrelation, but it is strongly rejected that the residuals are normally distributed. Therefore we use QMLE and the robust standard error estimator.



(a) Histogram of residuals

| Test                  | Statistic (p-value) |
|-----------------------|---------------------|
| Portmanteau test      | 0.98687 [0.4855]    |
| No-Heteroskedasticity | 1.8815 [0.0001]     |
| Normality test        | 72.736 [0.0000]     |
| RESET-test            | 0.63145 [0.7517]    |

(b) Misspecification test for the estimated VAR model



(c) Roots of the companion matrix

Figure 2: Misspecification and stability analysis

From here we simplify our model by removing statistically insignificant terms. A joint robust Wald test is conducted with null hypothesis  $\mathcal{H}_0 : \Pi_l = 0, l = 4, 5, \dots, 10$ . The test results in a p-value of approximately 0.66, which means, we cannot reject the null hypothesis and we omit these lags. We include dummies at critical points, from an economic standpoint, to account for observations that result in large residuals, which we do not believe are generated by a VAR process.

Tabel 2b presents the misspecification tests.<sup>6</sup> Our estimated model does not indicate autocorrelation or wrong functional form (both the RESET- and Portmantau / Ljung-

<sup>6</sup>Note that the p-values are reliable given that our model is well specified, the series are stationary and certain regularity conditions hold.

Box test have p-values larger than all conventional significant levels). However the test statistics does indicate heteroscedasticity and non-normally distributed errors. However this does not imply that our estimates are inconsistent.

From figure 2c we conclude that the inverse roots of the companion matrix are located within the complex unit circle. This is a sufficient condition for stability and stationarity, cf. subsubsection 1.2.1 given correct model specification.

### 1.4.2 Estimation results

The estimated parameters of the reduced VAR(12) model for the 427 observations between January 1990 and July 2025 is given in equation (1.9). (Included dummies<sup>7</sup>)

$$\begin{pmatrix} \Delta \text{Spread}_t \\ \Delta \log \text{CPI}_t \end{pmatrix} = \begin{pmatrix} 0.012 (0.016) \\ 0.001300 (0.000211) \end{pmatrix} + \begin{pmatrix} 0.331 (0.050) & 1.873 (4.593) \\ 0.000360 (0.000571) & 0.466 (0.071) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-1} \\ \Delta \log \text{CPI}_{t-1} \end{pmatrix} \\ + \begin{pmatrix} -0.120 (0.049) & -5.021 (4.733) \\ -0.000488 (0.000475) & -0.155 (0.066) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-2} \\ \Delta \log \text{CPI}_{t-2} \end{pmatrix} + \begin{pmatrix} 0.152 (0.047) & 1.195 (4.558) \\ 0.000432 (0.000486) & 0.097 (0.049) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-3} \\ \Delta \log \text{CPI}_{t-3} \end{pmatrix} \\ + \begin{pmatrix} 0.021 (0.052) & -1.945 (4.476) \\ -0.000053 (0.000458) & 0.196 (0.050) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-11} \\ \Delta \log \text{CPI}_{t-11} \end{pmatrix} + \begin{pmatrix} -0.036 (0.048) & -0.155 (4.231) \\ 0.0000658 (0.000513) & -0.160 (0.057) \end{pmatrix} \begin{pmatrix} \Delta \text{Spread}_{t-12} \\ \Delta \log \text{CPI}_{t-12} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}. \quad (1.9)$$

The estimated contemporaneous correlation between the two variables are

$$\hat{\rho} = \text{corr}(\Delta \text{Spread}, \Delta \log(\text{CPI})) \approx 0.116 \quad (1.10)$$

We check whether the correlation is significant. Under the null hypothesis,  $\hat{\rho}$  is  $N(0, T^{-1})$  distributed<sup>8</sup>. The the 5 pct. critical value is  $\frac{1.96}{\sqrt{T}} = \frac{1.96}{\sqrt{427}} \approx 0.095$ . We reject the null hypothesis, implying that the contemporaneous effect is significantly different from 0 - in line with conventional theory and our expectations regarding the contemporaneous relationship discussed earlier.

The interpretation of (1.10) is that a 1 pct. change in the consumer price index is associated with an increased interest rate spread of 0.116 (corresponding to a relatively higher long rate, or equivalently a relatively lower short rate). This suggests that both series may respond to one another, and may respond contemporaneously to common underlying shocks, such as monetary policy or expectation driven shocks.

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<sup>7</sup>The model includes the following dummies: 2008(10), 2008(11), 2008(12), 2022(7). Note that these occur in times of extraordinary insecurity (financial crisis and energy crisis).

<sup>8</sup>(p. 138 Nielsen, 2025)

### 1.4.3 Granger Causality

We test for Granger Causality between the interest rate spread and the consumer price index using the Wald-test. The null-hypothesis of no-Granger causality is given by

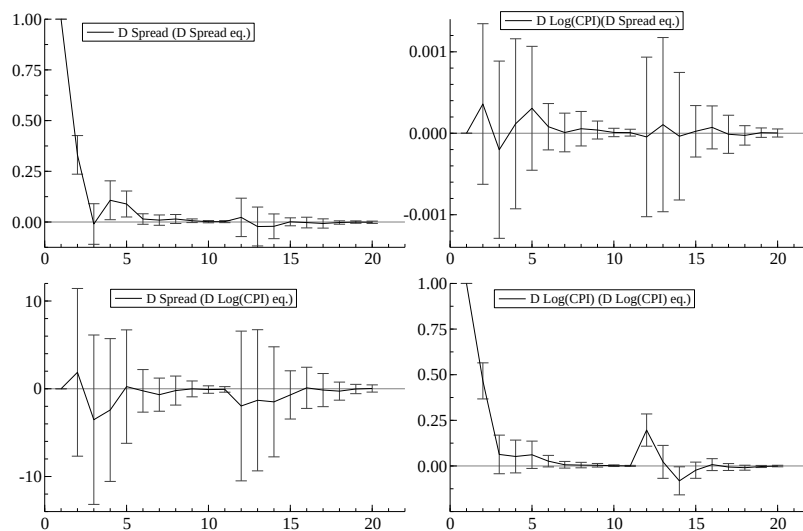
$$\begin{aligned}\mathcal{H}_{\Delta Spread \nrightarrow \Delta \log CPI} : \Pi_{12}^1 &= \Pi_{12}^2 = \Pi_{12}^3 = \Pi_{12}^{11} = \Pi_{12}^{12} = 0 \\ \mathcal{H}_{\Delta \log CPI \nrightarrow \Delta Spread} : \Pi_{21}^1 &= \Pi_{21}^2 = \Pi_{21}^3 = \Pi_{21}^{11} = \Pi_{21}^{12} = 0\end{aligned}\tag{1.11}$$

The corresponding robust Wald-test statistics are presented in (1.12). We cannot reject the null-hypothesis at any conventional level of significance in either case. From this we conclude that neither variable seem to Granger-causes the other at the monthly frequency. In other words, we cannot reject, that past values of the term spread do not help predict inflation, and vice versa.

$$\begin{aligned}W[\Delta Spread \nrightarrow \log(\Delta CPI)] &= 1.7334 \quad (p = 0.8847) \\ W[\log(\Delta CPI) \nrightarrow \Delta Spread] &= 1.7177 \quad (p = 0.8867)\end{aligned}\tag{1.12}$$

### 1.4.4 Impulse response

To investigate the model's dynamic structure, we calculate the impulse response function (see Figure 3). The impulse response function illustrates how shocks in one variable propagate through its own process over time and affect the other variable in the estimated VAR model.



**Figure 3: Unit impulse response**

From Figure 3 we note that the model suggests, that consumer prices are very weakly affected by shocks to the interest rate spread. There is considerable statistical uncertainty in these estimates; as a result, none of the estimates are significantly different from zero

at the 5% level. Qualitatively the same conclusion holds when observing the effects of shocks to the spread.

However, shocks to each process clearly affect that variable's own dynamic development, especially for the CPI, which strongly depends on its lagged value  $\Delta \log(\text{CPIAUCSL}_{t-1})$ . Additionally, most of the dynamic effects on the variables themselves dissipate after about three periods (months).

## 1.5 Conclusion and discussion

In this paper we estimate a reduced VAR(12) model to investigate the dynamic relationship between the interest rate spread and inflation. Our Granger-causality tests provide no evidence that past values of the interest rate spread help predict inflation, nor that past inflation helps predict the spread, at the monthly frequency. However, we cannot reject the presence of contemporaneous dependence between the two series: the estimated contemporaneous correlation is positive (approximately 0.12), indicating that periods with relatively high spreads tend to coincide with periods of relatively high inflation, and vice versa.

In totality, these results suggest that the interest rate spread and inflation do not exert a clear dynamic (lagged) influence on each other in our sample, but they may respond simultaneously to common underlying shocks. An Structural VAR model could be an alternative modelling approach. This however, requires making an assumption on the causal direction between the variables, which we would not advocate in this case. One of the main motivations for using a VAR model is precisely that the contemporaneous causal direction between spread and inflation is unclear.

Overall, we find that the dynamic (lagged) relationship between the variables is weak, even if economic theory might suggest a stronger link. These findings may reflect measurement error or the fact that central banks set short-term rates using information that may not be publicly available at the time of the decision.

## Part II: Assignment #4

### Transmissions from the US to the European Stock Market

## 2.1 Introduction

This paper studies fundamental characteristics of the European stock index, Stoxx 600. In addition we investigate the potential spill over effects from the American traded S&P500 index on Stoxx 600. To conduct this analysis we utilize the GARCH framework to model the conditional variance and mean of the within trading day return.

Later on we extend the model to a threshold GARCHX model and include lagged values of the within trading day return of the SP500 index. Specifically we estimate an ADL(2,1)-GARCH(1,1)X model using quasi ML estimation. Applying misspecification tests we conclude that the model does not indicate misspecification.

The empirical results point to a positive return spillover from the S&P500 to the Stoxx 600 index, which means that an increase in the within trading day return of the SP500 in the previous period has a positive impact on the within trading day return of the Stoxx 600 in the current period with an elasticity of about 0.2. However, we find a less clear volatility spillover as we estimate a borderline significant parameter with low economic significance.

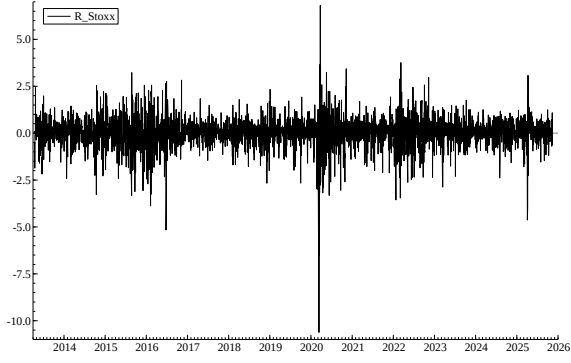
## 2.2 Presentation of data

In this paper we consider the within trading day returns of the European Stoxx 600 index, covering the period from April 15 2015 to November 13 2025. Furthermore, we use the within trading day returns of the American S&P500 index to determine whether or not this index is helpful in predicting the conditional variance and mean of the Stoxx 600 index. We define the within trading day returns as follows

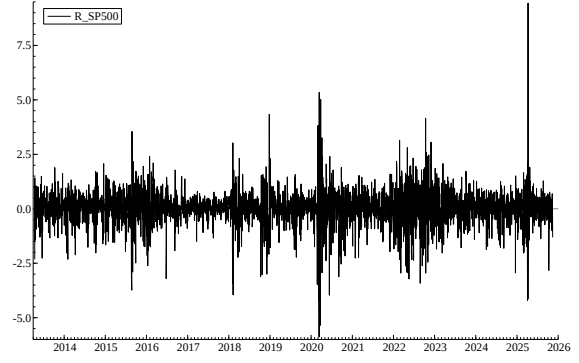
$$R\_Stoxx_t = 100 \cdot [\text{Log}(\text{Close\_Stoxx}_t) - \text{Log}(\text{Open\_Stoxx}_t)] \quad (2.1)$$

$$R\_SP500_t = 100 \cdot [\text{Log}(\text{Close\_SP500}_t) - \text{Log}(\text{Open\_SP500}_t)] \quad (2.2)$$

From figure 4 it is clear that the (within trading day) returns of the two stock indexes exhibit well known stylized characteristics of financial time series. There is clear volatility clustering and fluctuations orders of magnitude larger than the standard fluctuations. This was for instance the case during the Covid 19 pandemic, but also later in the context of the Russian invasion of Ukraine and the following the rise in energy prices. Therefore the natural modelling framework will be the ARCH-family of time series models. In the econometric analysis that follow we also find additional standard features of stock returns: asymmetric effects of returns on future volatility, and no statistically or economically significant predictability in the conditional mean returns.



(a) Stoxx 600



(b) SP500

Figure 4: Characterization of the within trading day returns

## 2.3 Econometric theory

### 2.3.1 GARCH and TGARCH

Section 2.2 argues that the data exhibits substantial volatility clustering. GARCH models try to take this into account by modelling the time-varying variance as an autoregressive (or ADL) process. Specifically, we model the conditional time varying variance and the conditional mean. We consider the threshold GARCH, TGARCH(p,q), process given by

$$y_t = x_t' \beta + \epsilon_t \quad (2.3)$$

$$\epsilon_t = \sigma_t z_t, \quad z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} (0, 1) \quad (2.4)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \tau_i \epsilon_{t-i}^2 I(\epsilon_{t-i} < 0) + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad \alpha_i, \beta_i, \tau_i \geq 0 \forall i, j, \omega > 0 \quad (2.5)$$

, where  $\mathcal{I}_{t-1} = \{x_1, y_{t-1}, x_{t-1}, y_{t-2}, x_{t-2}, \dots\}$  defines the information set. The conditional mean  $x_t' \beta$  potentially contains all or some of the following: lagged covariates, other explanatory variables and dummies. The residuals  $\epsilon_t$  are defined by the product between the conditional standard deviation  $\sigma_t$  and the independently and identically distributed innovations  $z_t$ . The innovations take on an arbitrary distribution with mean zero and variance 1. From (2.5) we can interpret the modelling of the conditional variance. Shocks to the process will propagate through the conditional variance determined by the coefficients. Including exogenous variables in eq. (2.5) the unconditional variance for a symmetric distributed  $z_t$  is given by:

$$\sigma^2 = \frac{\omega + \delta}{1 - \sum_{i=1}^p \alpha_i - \frac{1}{2} \sum_{i=1}^p \tau_i - \sum_{j=1}^q \beta_j} \quad (2.6)$$

where  $\delta$  contains the coefficients on the exogenous variables. From (2.6) we note that the unconditional variance exists for

$$\sum_{i=1}^p \alpha_i + \frac{1}{2} \sum_{i=1}^p \tau_i + \sum_{j=1}^q \beta_j < 1 \quad (2.7)$$

, which is a sufficient stationarity condition for the process. To investigate asymmetric effects in the GARCH framework, we extend the model. Namely we utilize the TGARCH and TGARCH X models, which allow for asymmetric volatility responses to innovations. Specifically this alters (2.5) and we have the following general ADL(A,B)-TGARCH(p,q)-X(w) model specification given by,

$$R\_Stoxx_t = \sum_{a=1}^A \phi_a R\_Stoxx_{t-a} + \sum_{b=1}^B \mu_b R\_SP500_{t-b} \quad (2.8)$$

$$\epsilon_t = \sigma_t z_t, \quad z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} (0, 1) \quad (2.9)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \tau_i \epsilon_{t-i}^2 I(\epsilon_{t-i} < 0) + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 + \sum_{k=1}^w \zeta_k R\_SP500_{t-k} \quad (2.10)$$

Note that for most practical purposes, it is sufficient to estimate a GARCH(1,1) model since  $\sigma_t^2$  exhibit a long lasting memory. The conditional variance is required to be non-negative and therefore we impose the following restrictions on the estimated parameters. In line with (2.5) we require  $\alpha_i, \beta_j, \zeta_k, \tau_i \geq 0 \forall i, j, k$  and  $\omega > 0$ . Note that these restrictions imply difficulties when trying to test the parameters in the conditional variance equation for statistical significance, either because 0 lies outside the parameter space or because the test statistics follow non-standard distributions.

### 2.3.2 (Quasi) Maximum Likelihood Estimation

To estimate the parameters of the model will need to specify a distributional shape of  $\epsilon_t$ . Cf our earlier discussion we want to allow for a larger proportion of extreme observations (compared to a normal distribution), and we therefore choose the students t-distribution. Formally we let  $z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} t(0, 1, \nu)$  or equivalently  $\epsilon_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} t(0, \sigma_t^2, \nu)$ , where  $\nu$  is the degrees of freedom. By scaling the standard t-distribution with  $\sqrt{(\nu-2)/\nu}$  we normalize so that  $Var(z_t) = 1$  for  $\nu > 2$ . We obtain the following scaled students  $t(\nu)$  log likelihood function.

$$\log L_T(\theta) = \sum_{t=1}^T \log \ell_t(\theta) = \sum_{t=1}^T \log \left[ \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\pi(\nu-2)\sigma_t^2(\theta)}} \left( 1 + \frac{(y_t - \mu_t(\theta))^2}{(\nu-2)\sigma_t^2(\theta)} \right)^{-\frac{\nu+1}{2}} \right] \quad (2.11)$$

The (quasi) maximum likelihood estimator is then defined as

$$\hat{\theta}_{QMLE} = \arg \max_{\theta \in \Theta} \log L_T(\theta) \quad (2.12)$$

Under certain high level assumptions<sup>9</sup> and given that the score  $s_t(\theta_0) = \frac{\partial}{\partial \theta} \ell_t(\theta)|_{\theta=\theta_0}$  satisfies  $\mathbb{E}[s_t(\theta_0)] = 0$  the estimator is consistent such that  $\hat{\theta}_{QMLE} \xrightarrow{p} \theta_0$ , where  $\theta_0 \in \Theta$  is the population parameters identifying the true process, and asymptotically normal such that

$$\sqrt{T}(\hat{\theta}_{QMLE} - \theta_0) \xrightarrow{d} N(0, \Omega_{\theta}^*(\theta_0)), \quad T \rightarrow \infty \quad (2.13)$$

where the asymptotic variance is a function of the true parameter  $\theta_0 \in \Theta$ , and can be shown to have the formula

$$\Omega_{\theta}^*(\theta_0) = \mathcal{I}(\theta_0)^{-1} \mathcal{J}(\theta_0) \mathcal{I}(\theta_0)^{-1} \quad (2.14^{10})$$

In the case of the model distribution correctly specifying the distribution of the data-generating process  $\mathcal{I} = \mathcal{J}$  and  $\Omega_{\theta}^*(\theta_0) = \mathcal{I}(\theta_0)^{-1}$  and the presented estimator is efficient.

### 2.3.3 Misspecification

We test for misspecifications of the model, specifically testing for no-ARCH effects and autocorrelation. The no-ARCH test is carried out by regressing the squared residuals on lagged values of the squared residuals<sup>11</sup>.

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \gamma_2 \hat{\varepsilon}_{t-2}^2 + \cdots + \gamma_p \hat{\varepsilon}_{t-p}^2 + \text{error} \quad (2.15)$$

The null hypothesis of no ARCH effects is

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_p = 0 \quad (2.16)$$

and is tested using an LM test statistic, which under the null follows a  $\chi^2(p)$  distribution. We also employ a normality test to test the the Gaussian MLE. See table 2 for the null hypothesis and the distribution of the tests for no-autocorrelation.

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<sup>9</sup>See Nielsen (2025), Assumption 3.1

<sup>10</sup>Where  $\mathcal{I}$  is the fisher-information  $\mathcal{I}(\theta_0) = -\mathbb{E} \left[ \frac{\partial^2 \log \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right]$  and  $\mathcal{J}(\theta_0) = \mathbb{E} \left[ \frac{\partial \log \ell_t(\theta_0)}{\partial \theta} \frac{\partial \log \ell_t(\theta_0)}{\partial \theta'} \right]$

<sup>11</sup>Nielsen (2025), section 12.3

<sup>11</sup>See section 2.3 "Formulation and Misspecification Testing," Nielsen (2025) for an explanation of the degrees of freedom for the test statistics asymptotic distributions.

| Test                          | Null hypothesis                | Asymptotic distribution<br>(under $\mathcal{H}_0$ ) |
|-------------------------------|--------------------------------|-----------------------------------------------------|
| No-ARCH test <sup>a</sup>     | no ARCH effects                | $\chi^2(q)$                                         |
| Portmanteau test <sup>b</sup> | no autocorrelation             | $\chi^2(h - p)$                                     |
| Normality test <sup>c</sup>   | normally distributed residuals | $\chi^2(2)$                                         |

<sup>a</sup> Nielsen (2025), p. 273.

<sup>b</sup> Slides 5: Model Formulation and Misspecification Testing.

<sup>c</sup> Nielsen (2025), p. 44.

**Table 2: Overview of misspecification tests and their null hypotheses**

## 2.4 Empirical results

### 2.4.1 Model selection and misspecification tests

In order to model the Stoxx 600 index and characterise the conditional mean and variance, we estimate a baseline AR(0) – TGARCH(1, 1)-model in accordance with (2.9)-(2.10).<sup>12</sup> Note for any initial general autoregressive order, a joint robust Wald test implies that we cannot reject the null hypothesis, that all lagged variables are insignificantly different from zero, at any conventional significance level. This implies that the past returns have no linear predictability regarding the current returns, as expected.

The estimated model appears well-specified. As table 3 (columns 2-5) shows, we cannot reject the null of no ARCH effects and autocorrelation.

| Test             | TGARCH (t)       | TGARCH X (t)     | TGARCH X (t)*   | TGARCH X (t)**   |
|------------------|------------------|------------------|-----------------|------------------|
| No ARCH test     | 0.39284 (0.6752) | 0.55601 (0.5735) | 1.0579 (0.3474) | 0.77001 (0.4632) |
| Portmanteau test | 68.828 (0.1355)  | 65.233 (0.2123)  | 54.143 (0.0820) | 49.361 (0.1236)  |

p values in parentheses. All models assume student-t distributed innovations.

\* Sample 2013-04-16 to 2019-12-31. \*\* Sample 2019-12-31 to 2025-11-13.

**Table 3: Misspecification tests for the TGARCH models**

To examine whether within trading day returns of the SP500 index help predict within trading day returns of the Stoxx 600 index, we extend model to an ALD(0,1)-TGARCH(1,1) X model, as described in Section 2.3.1. Table 3 (column 3) shows the misspecification test regarding the TGARCH X model and the model does not indicate misspecification.

<sup>12</sup>We also estimate the parameters based on  $z_t$  being conditionally standard normal distributed. The null of normality is strongly reject, and hence we carry out the main analysis as described in section 2.3 using the t-distribution. Note that we only require  $\mathbb{E}[\frac{\partial}{\partial \theta} L_t(\theta) | \theta = \theta_0] = 0$  along with the high level assumptions mentioned in section 2.3 for asymptotically normally distributed estimates. However the distributional choice still affects efficiency of the estimator.

### 2.4.2 Estimation results - Baseline TGARCH

Table 4 (column 1) reports the estimates from the baseline TGARCH model for the within trading day returns of the STOXX 600 index. The constant for the conditional mean is estimated at 0.04975, implying an average within trading day return of approximately 5%. The conditional variance constant,  $\omega = 0.03136$ , is positive and statistically significant but small relative to the strong persistence component of the models conditional variance.:  $\beta = 0.81713$ . The coefficient  $\alpha$  is restricted to be zero (or above) with reported value  $\alpha = 0.00000$ , suggesting that positive innovations have little impact on the next periods volatility - beyond what is captured by the lagged conditional variance. In contrast, the threshold parameter  $\tau = 0.25648$  is positive and statistically significant, indicating that negative innovations lead to a substantial increase in conditional variance. Note as well that  $\alpha + \beta + \frac{1}{2}\tau = 0.945$ , is below 1, implying that volatility is highly persistent yet covariance-stationary. Overall, the results provide strong evidence of asymmetric volatility dynamics and are consistent with the existing literature on stock returns.

| Parameter                          | TGARCH          | TGARCH X        | TGARCH X*       | TGARCH X**      |
|------------------------------------|-----------------|-----------------|-----------------|-----------------|
| Constant                           | 0.04975 (0.011) | 0.03611 (0.010) | 0.02904 (0.014) | 0.04715 (0.016) |
| $R_{SP500_{t-1}}$                  |                 | 0.16905 (0.017) | 0.16487 (0.028) | 0.17268 (0.022) |
| $R_{SP500_{t-1}}^2$                |                 | 0.03360 (0.011) | 0.03397 (0.019) | 0.04099 (0.016) |
| $\omega$                           | 0.03136 (0.007) | 0.02927 (0.006) | 0.02844 (0.009) | 0.03097 (0.009) |
| $\alpha$                           | 0.00000         | 0.00000         | 0.00000         | 0.02432 (0.026) |
| $\beta$                            | 0.81713 (0.024) | 0.78703 (0.025) | 0.79601 (0.046) | 0.75376 (0.038) |
| $\tau$ (Threshold)                 | 0.25648 (0.048) | 0.25874 (0.043) | 0.25163 (0.066) | 0.25364 (0.063) |
| student t df                       | 5.75327 (0.555) | 6.04170 (0.609) | 6.42986 (0.973) | 5.59353 (0.744) |
| $\alpha + \frac{1}{2}\tau + \beta$ | 0.945           | 0.916           | 0.922           | 0.905           |
| Observations                       | 3256            | 3255            | 1734            | 1522            |

Robust standard errors in parentheses. \* 2013-04-16 to 2019-12-31. \*\* 2019-12-31 to 2025-11-13.

**Table 4: Estimation results for TGARCH models**

### 2.4.3 Estimation results - Extended TGARCH X

The extended model (see column 2 table 4) now includes the lagged SP500 within trading day returns in the mean equation and the squared SP500 within trading day returns in the variance equation. The estimate for  $R_{SP500_{t-1}}$  shows an economically significant return spillover from SP500 to Stoxx 600 (0.16905). In practice this means that a 1 pp. increase in the within trading day returns of the SP500 in period  $t - 1$  entails a positive increase to the within trading day returns of the Stoxx 600 on 0.17 pp. in period  $t$ .

The volatility spillover from SP500 to Stoxx 600 is borderline significant and economically modest. A 1 pp. shock to the squared within trading day returns of the SP500 in period  $t-1$  results in a 0.036 pp. increase in the conditional variance of  $R\_Stoxx$  in period  $t$ . The remaining parameters are close to the baseline, and are interpreted similarly to the baseline model, with  $\alpha$  being totally insignificant. This indicates that the conditional volatility of the within trading day returns of the European stock index, primarily can be described by its own past shocks, and past volatility, potentially with some small level of spillover from the SP500.

To determine whether or not the spillover effects from the American stock market to the European stock market are constant over time, we estimate the TGARCH X model for two sub-periods. The first runs from 2013 to the end of 2019 (pre covid-19), and the second covers the period from the end of 2019 to 2025 (during and post Covid-19). As seen in table 3, the models remain well specified. The results (see table 4) show that the return spillover is remarkably stable, with only slight deviations from the full sample estimate. The volatility spillover effect varies more from the first sub-period to the second. This indicates that the volatility spillover effect may have become stronger in the later years, although the level remains low. Overall, the sub-period analysis confirms the results we have found previously.

## 2.5 Conclusion

The empirical analysis using our final model TGARCH-X shows that the spillover effects from the American stock market to the European stock market finds a substantial and significant return spillover of 0.16905 and a less impactful, and less significant volatility spillover effect 0.03360. In practise this implies that the within trading day returns of the American SP500 index, yesterday, affects the within trading day returns of the European Stoxx 600 index today. In contrast we find that the volatility of the Stoxx 600 index is primarily driven by persistence in the volatility  $\beta = 0.79$  and previous negative shocks  $\tau = 0.26$  to the series itself, and is not affected substantially by the SP500 index.

These findings are inline with other literature suggesting a non-negligible return spillover from the leading US stock indexes to Europe, but also other parts for the world. The findings regarding the volatility spillover are also inline with other literature finding that past volatility is more impactful to the present volatility than the volatility originating from other markets (Dimpfl, 2011).

## Part III: Theoretical assignment

## 5.1 A Misspecified Autoregression

(1) The data generating process is given as:

$$y_t = \theta_0 y_{t-1} + \rho_0 y_{t-2} + \eta_t \quad (5.1)$$

, with  $\eta_t \stackrel{iid}{\sim} N(0, \omega^2)$ . The DGP is stationary if  $-2 < \theta_0 + \rho_0 - 1 < 0$ . We need to derive an expression for:

$$\gamma_j = Cov(y_t, y_{t-j}), \quad j = 1, 2, \dots \quad (5.2)$$

Since there is no constant in (5.1), the DGP is assumed to be centred. Combined with stationarity, this means:  $E[y_t] = E[y_{t-j}] = 0$  implying

$$Cov(y_t, y_{t-j}) = E[y_t y_{t-j}] \quad (5.3)$$

We now need to derive  $E[y_t y_{t-j}] = \gamma_j$ . We are given  $\gamma_0 = V(y_t) = 1$ : We obtain  $\gamma_1$  by multiplying the DGP with  $y_{t-1}$  and take expectations:

$$E[y_t y_{t-1}] = \theta_0 E[y_{t-1}^2] + \rho_0 E[y_{t-2} y_{t-1}] + E[\eta_t y_{t-1}] \quad (5.4)$$

where

$$E[\eta_t y_{t-1}] = 0, \text{ since } \eta_t \text{ is not affected by the past,} \quad (5.5)$$

$$E[y_t y_{t-1}] = E[y_{t-2} y_{t-1}] = \gamma_1, \quad E[y_{t-1}^2] = \gamma_0 = 1 \quad (5.6)$$

This is inserted into (5.4).

$$E[y_t y_{t-1}] = \theta_0 E[y_{t-1}^2] + \rho_0 E[y_{t-2} y_{t-1}] + E[\eta_t y_{t-1}] \Leftrightarrow \gamma_1 = \theta_0 + \rho_0 \gamma_1 \quad (5.7)$$

$$\Leftrightarrow \gamma_1 = \frac{\theta_0}{1 - \rho_0} \quad (5.8)$$

If we follow the same steps, but instead multiply the DGP with  $y_{t-2}$  and take expectations we get:

$$E[y_t y_{t-2}] = \theta_0 E[y_{t-1} y_{t-2}] + \rho_0 E[y_{t-2}^2] + E[\eta_t y_{t-2}] \quad (5.9)$$

where

$$E[\eta_t y_{t-2}] = 0, \quad E[y_t y_{t-2}] = \gamma_2, \quad E[y_{t-2}^2] = \gamma_0 = 1, \quad E[y_{t-1} y_{t-2}] = \gamma_1 \quad (5.10)$$

$$\gamma_2 = \theta_0 \gamma_1 + \rho_0 \quad (5.11)$$

We again follow the same steps, but instead multiply the DGP with  $y_{t-3}$  and take expectations we get:

$$E[y_t y_{t-3}] = \theta_0 E[y_{t-1} y_{t-3}] + \rho_0 E[y_{t-2} y_{t-3}] + E[\eta_t y_{t-3}] \quad (5.12)$$

$$\gamma_3 = \theta_0 \gamma_2 + \rho_0 \gamma_1 \quad (5.13)$$

For the remaining cases  $j = 3, 4, \dots$  we notice that

$$E[y_t y_{t-j}] = \theta_0 E[y_{t-1} y_{t-j}] + \rho_0 E[y_{t-2} y_{t-j}] + E[\eta_t y_{t-j}] \quad (5.14)$$

$$\gamma_j = \theta_0 \gamma_{j-1} + \rho_0 \gamma_{j-2} \quad (5.15)$$

Combined we have the covariance of order  $j$  given by

$$\gamma_j = \begin{cases} \frac{\theta_0}{1-\rho_0}, & \text{for } j = 1 \\ \theta_0 \gamma_1 + \rho_0, & \text{for } j = 2 \\ \theta_0 \gamma_{j-1} + \rho_0 \gamma_{j-2}, & \text{for } j = 3, 4, \dots \end{cases} \quad (5.16)$$

(2) Under the assumed model the conditional distribution is normal, the error terms are i.i.d and the density function is given below. Remember that the assumed distribution implies  $\epsilon_t = y_t - \theta y_{t-1}$ . From this we get

$$f(y_t | y_{t-1}, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{\epsilon_t^2}{2\sigma^2}\right] = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(y_t - \theta y_{t-1})^2}{2\sigma^2}\right] \quad (5.17)$$

The likelihood contribution from observation  $t \in \{1, 2, \dots, T\}$  is given by,

$$\ell_t(\theta, \sigma^2) = f(y_t | y_{t-1}, \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(y_t - \theta y_{t-1})^2}{2\sigma^2}\right] \quad (5.18)$$

and to find the log likelihood contributions, we take the log of the likelihood contributions.

$$\log \ell_t(\theta, \sigma^2) = \log \left[ \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left[-\frac{(y_t - \theta y_{t-1})^2}{2\sigma^2}\right] \right] \quad (5.19)$$

$$= \log \left[ \frac{1}{\sqrt{2\pi\sigma^2}} \right] + \log \left[ \exp\left[-\frac{(y_t - \theta y_{t-1})^2}{2\sigma^2}\right] \right] \quad (5.20)$$

$$= \log \left[ (2\pi\sigma^2)^{-\frac{1}{2}} \right] - \frac{(y_t - \theta y_{t-1})^2}{2\sigma^2} \quad (5.21)$$

$$= -\frac{1}{2} \log[2\pi\sigma^2] - \frac{(y_t - \theta y_{t-1})^2}{2\sigma^2} \quad (5.22)$$

The error terms are assumed to be independent distributed and the likelihood for the full sample is therefore the product of the individual likelihood contributions or the the sum

of the log likelihood contributions. To find the log likelihood function we therefore sum all the likelihood contributions.

$$\log L_T(\theta, \sigma^2 \mid y_0) = \sum_{t=1}^T -\frac{1}{2} \log[2\pi\sigma^2] - \frac{(y_t - \theta y_{t-1})^2}{2\sigma^2} \quad (5.23)$$

$$= -\frac{T}{2} \log[2\pi\sigma^2] - \frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \theta y_{t-1})^2 \quad (5.24)$$

We have now derived the log likelihood function for the statistical model  $y_t = \theta y_{t-1} + \epsilon_t$ , for  $t = 1, 2, \dots, T$  conditioned on  $y_0$ .

**Derive score contribution.** We derive the score contribution of observation  $t$  with respect to  $\theta$  by differentiating the log likelihood contribution of observation  $t$  with respect to  $\theta$  and applying the chain rule from calculus.

$$s_t(\theta, \sigma^2) = \frac{\partial \log \ell_t(\theta, \sigma^2)}{\partial \theta} \quad (5.25)$$

$$= \frac{\partial}{\partial \theta} \left( -\frac{1}{2} \log[2\pi\sigma^2] - \frac{(y_t - \theta y_{t-1})^2}{2\sigma^2} \right) \quad (5.26)$$

$$= 0 - \frac{1}{2\sigma^2} \cdot 2 \cdot (y_t - \theta y_{t-1})(-y_{t-1}) \quad (5.27)$$

$$= \frac{1}{\sigma^2} y_{t-1} (y_t - \theta y_{t-1}) \quad (5.28)$$

We calculate the expected value of the score contribution  $s_t(\theta, \sigma^2)$  evaluated at the population parameter values  $\theta_0, \sigma_0^2$ :

$$s_t(\theta_0, \sigma_0^2) = \frac{1}{\sigma_0^2} y_{t-1} (y_t - \theta_0 y_{t-1}) \quad (5.29)$$

$$E[s_t(\theta_0, \sigma_0^2)] = E \left[ \frac{1}{\sigma_0^2} y_{t-1} (y_t - \theta_0 y_{t-1}) \right] = \frac{1}{\sigma_0^2} E[y_t y_{t-1}] - \frac{1}{\sigma_0^2} \theta_0 E[y_{t-1}^2] \quad (5.30)$$

Because  $E[y_t] = E[y_{t-j}] = 0$ , it is the case that  $E[y_{t-1}^2]$  is equal to the variance ( $\gamma_0$ ). Furthermore  $E[y_{t-1} y_t]$  is equal to  $\gamma_1$  and we insert this into the equation. At the same time recall from (1) that  $\gamma_0 = 1$  and  $\gamma_1 = \frac{\theta_0}{1-\rho_0}$ .

$$= \frac{1}{\sigma_0^2} \gamma_1 - \frac{\theta_0}{\sigma_0^2} \gamma_0 = \frac{1}{\sigma_0^2} \cdot \frac{\theta_0}{1-\rho_0} - \frac{1}{\sigma_0^2} \cdot \theta_0 = \frac{\theta_0}{\sigma_0^2} \left[ \frac{\rho_0}{1-\rho_0} \right] \quad (5.31)$$

, for  $\rho_0 \neq 1$ . For the quasi maximum likelihood estimator to be consistent, it is a condition that  $E[s_t(\theta_0, \sigma_0^2)] = 0$ <sup>13</sup>. In this case when  $\rho_0$  is different from 0, the condition is not satisfied and the QMLE does not converge to the true parameter in probability (ie. the estimator is inconsistent). This result arises because the AR(1)-model is misspecified. The omitted  $y_{t-2}$  is correlated with  $y_{t-1}$ , which implies that we have autocorrelation in the residuals, which is in direct conflict with the assumed independent distributed error terms.

**(3)** The score with respect to  $\theta$  is given by the sum of the score contributions found in the previous question. To derive the QMLE of  $\theta$  we set the score equal to 0.

$$S_T(\theta) = \sum_{t=1}^T \frac{1}{\sigma^2} y_{t-1}(y_t - \hat{\theta} y_{t-1}) = 0 \Leftrightarrow \frac{1}{\sigma^2} \sum_{t=1}^T y_{t-1}(y_t - \hat{\theta} y_{t-1}) = 0 \quad (5.32)$$

$$\Leftrightarrow \sigma^2 \cdot \frac{1}{\sigma^2} \sum_{t=1}^T y_{t-1}(y_t - \hat{\theta} y_{t-1}) = 0 \cdot \sigma^2 \Leftrightarrow \sum_{t=1}^T y_{t-1}(y_t - \hat{\theta} y_{t-1}) = 0 \quad (5.33)$$

$$\Leftrightarrow \sum_{t=1}^T y_{t-1} y_t - \hat{\theta} \sum_{t=1}^T y_{t-1}^2 = 0 \Leftrightarrow \sum_{t=1}^T y_{t-1} y_t = \hat{\theta} \sum_{t=1}^T y_{t-1}^2 \quad (5.34)$$

$$\Leftrightarrow \hat{\theta} = \frac{\sum_{t=1}^T y_{t-1} y_t}{\sum_{t=1}^T y_{t-1}^2} \quad (5.35)$$

The estimator for  $\theta$  is now derived and note it is the OLS estimator. Under a Law of large numbers (since the true DGP follows a stationary and weakly dependent process) the numerator and denominator of the estimator will converge to the population moments respectively for  $T \rightarrow \infty$ .

$$\hat{\theta} = \frac{\sum_{t=1}^T y_{t-1} y_t}{\sum_{t=1}^T y_{t-1}^2} \xrightarrow{p} \frac{E[y_{t-1} y_t]}{E[y_{t-1}^2]} = \frac{\gamma_1}{\gamma_0} \quad (5.36)$$

We know from previous questions that  $\gamma_0 = 1$  and  $\gamma_1 = \frac{\theta_0}{1-\rho_0}$ , which implies that  $\hat{\theta}$  is converging to:

$$\text{plim } \hat{\theta} = \frac{\gamma_1}{\gamma_0} = \frac{\frac{\theta_0}{1-\rho_0}}{1} = \frac{\theta_0}{1-\rho_0} \quad (5.37)$$

This shows explicitly that the estimator is not consistent, since consistency requires that  $\hat{\theta} \xrightarrow{p} \theta_0$  as  $T \rightarrow \infty$ , cf. Theorem 3.1<sup>14</sup>. The asymptotic bias of the estimator is given by

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<sup>13</sup>section 3.3.2., Nielsen (2025)

<sup>14</sup>Nielsen (2025)

the probability limit of  $\hat{\theta} - \theta_0$  as  $T \rightarrow \infty$ , and in this case it is given as follows.

$$\text{plim } \hat{\theta} - \theta_0 = \frac{\theta_0}{1 - \rho_0} - \theta_0 = \frac{\theta_0 - (1 - \rho_0)\theta_0}{1 - \rho_0} = \frac{\rho_0\theta_0}{1 - \rho_0} \quad (5.38)$$

(4) To fill out the table, we calculate the asymptotic bias of the estimator by inserting the given values of  $\rho_0$  and  $\theta_0$ .

$$\text{Bias}(\hat{\theta}) = \frac{\rho_0\theta_0}{1 - \rho_0} = \frac{(-0.4)(-0.5)}{1 - (-0.5)} \approx 0.133, \text{ if } \theta_0 = -0.4 \text{ and } \rho_0 = -0.5 \quad (5.39)$$

The table below reports the asymptotic biases, calculated as shown above.

|                   | $\rho_0 = -0.5$ | $\rho_0 = 0$ | $\rho_0 = 0.5$ |
|-------------------|-----------------|--------------|----------------|
| $\theta_0 = -0.4$ | 0.133           | 0            | -0.4           |
| $\theta_0 = 0$    | 0               | 0            | 0              |
| $\theta_0 = 0.4$  | -0.133          | 0            | 0.4            |

## 5.2 A Model with Two Outcomes

(1) The log likelihood contribution of a stochastic variables  $y_t$  is simply the probability density function as a function of the parameters that define the probability density. We have the likelihood contribution of observation  $t$  conditional on  $(y_t, x_t)$  given by

$$\ell_t(\beta) = p_t^{y_t} (1 - p_t)^{1-y_t} \quad (5.40)$$

$$= \left( \frac{1}{1 + \exp(-x'_t \beta)} \right)^{y_t} \left( 1 - \left( \frac{1}{1 + \exp(-x'_t \beta)} \right) \right)^{1-y_t} \quad (5.41)$$

$$= \left( \frac{1}{1 + \exp(-(a + bz_t))} \right)^{y_t} \left( 1 - \left( \frac{1}{1 + \exp(-(a + bz_t))} \right) \right)^{1-y_t} \quad (5.42)$$

Taking the log of (5.40) we have the log likelihood contribution (of observation  $t$ ) given by

$$\log \ell_t(\beta) = y_t \log(p_t) + (1 - y_t) \log(1 - p_t) \quad (5.43)$$

$$= -y_t \log(1 + \exp(-x'_t \beta)) - (1 - y_t)(x'_t \beta + \log(1 + \exp(-x'_t \beta))) \quad (5.44)$$

$$= -y_t \log(1 + \exp(-(a + bz_t))) - (1 - y_t)(a + bz_t + \log(1 + \exp(-(a + bz_t)))) \quad (5.45)$$

Since we assume independence across observations the simultaneous density function is simply the product of the marginal densities. This implies that the log likelihood function is the sum of the log likelihood contributions (these are equal to the marginal densities). The log-likelihood function is given by

$$\log L_t(\beta) = \sum_{t=1}^T \log \ell_t(\beta) = \sum_{t=1}^T \left[ y_t \log(p_t) + (1 - y_t) \log(1 - p_t) \right] \quad (5.46)$$

We have now written down the (log) likelihood contributions  $\log \ell_t(\beta)$  and the log-likelihood function  $L_t(\beta)$ .

(2) By direct calculation and by applying vector derivatives we have

$$\frac{\partial}{\partial \beta}(p_t) = \frac{\partial}{\partial \beta} \left( \frac{1}{1 + \exp(-x'_t \beta)} \right) \quad (5.47)$$

$$= \begin{bmatrix} \frac{\partial}{\partial a} \left( \frac{1}{1 + \exp(-x'_t \beta)} \right) \\ \frac{\partial}{\partial b} \left( \frac{1}{1 + \exp(-x'_t \beta)} \right) \end{bmatrix} \quad (5.48)$$

We apply the chain rule and get

$$\begin{bmatrix} -\frac{1}{(1+\exp(-x'_t\beta))^2} \exp(-x'_t\beta) \frac{\partial}{\partial a}(-x'_t\beta) \\ -\frac{1}{(1+\exp(-x'_t\beta))^2} \exp(-x'_t\beta) \frac{\partial}{\partial b}(-x'_t\beta) \end{bmatrix} = \begin{bmatrix} -\frac{1}{(1+\exp(-x'_t\beta))^2} \exp(-x'_t\beta) \frac{\partial}{\partial a}(-x'_t\beta) \\ -\frac{1}{(1+\exp(-x'_t\beta))^2} \exp(-x'_t\beta) \frac{\partial}{\partial b}(-x'_t\beta) \end{bmatrix} \quad (5.49)$$

$$= \begin{bmatrix} \frac{\exp(-x'_t\beta)}{(1+\exp(-x'_t\beta))^2} \\ \frac{\exp(-x'_t\beta)z_t}{(1+\exp(-x'_t\beta))^2} \end{bmatrix} \quad (5.50)$$

$$= \begin{bmatrix} \frac{1}{1+\exp(-x'_t\beta)} \cdot \frac{\exp(-x'_t\beta)}{1+\exp(-x'_t\beta)} \\ \frac{1}{(1+\exp(-x'_t\beta))} \frac{\exp(-x'_t\beta)}{1+\exp(-x'_t\beta)} z_t \end{bmatrix} \quad (5.51)$$

In order to see how we can reduce (5.51) it can be helpful to realise that

$$1 - p_t = 1 - \frac{1}{1 + \exp(-x'_t\beta)} = \frac{1 + \exp(-x'_t\beta) - 1}{1 + \exp(-x'_t\beta)} = \frac{\exp(-x'_t\beta)}{1 + \exp(-x'_t\beta)} \quad (5.52)$$

Therefore inserting (5.52) into (5.51) the calculations are complete

$$\begin{bmatrix} \frac{1}{1+\exp(-x'_t\beta)} \cdot \frac{\exp(-x'_t\beta)}{1+\exp(-x'_t\beta)} \\ \frac{1}{(1+\exp(-x'_t\beta))} \frac{\exp(-x'_t\beta)}{1+\exp(-x'_t\beta)} z_t \end{bmatrix} = \begin{bmatrix} p_t(1 - p_t) \\ p_t(1 - p_t)z_t \end{bmatrix} \quad (5.53)$$

The desired result has now been derived. To recap, we have shown that

$$\begin{bmatrix} \frac{\partial p_t}{\partial a} \\ \frac{\partial p_t}{\partial b} \end{bmatrix} = \begin{bmatrix} p_t(1 - p_t) \\ p_t(1 - p_t)z_t \end{bmatrix} \quad (5.54)$$

or completely equivalently that  $\frac{\partial p_t}{\partial a} = p_t(1 - p_t)$  and  $\frac{\partial p_t}{\partial b} = p_t(1 - p_t)z_t$ .

**Deriving the score.** To derive the score contributions with respect to  $\beta$  we - as the first step - apply the chain rule and the product rule to obtain:

$$\frac{\partial \log \ell_t(\beta)}{\partial \beta} = \frac{1}{\ell_t(\beta)} \frac{\partial \ell_t(\beta)}{\partial \beta} \quad (5.55)$$

$$= \frac{1}{\ell_t(\beta)} \left( y_t(p_t)^{y_t-1} \frac{\partial p_t}{\partial \beta} (1 - p_t)^{1-y_t} + p_t^{y_t} (1 - y_t)(1 - p_t)^{-y_t} \frac{\partial(1 - p_t)}{\partial \beta} \right) \quad (5.56)$$

$$= \frac{1}{\ell_t(\beta)} \left( \frac{\partial p_t}{\partial \beta} \left[ (y_t p_t^{y_t-1} (1 - p_t)^{1-y_t} - p_t^{y_t} (1 - y_t)(1 - p_t)^{-y_t}) \right] \right) \quad (5.57)$$

From here we utilize the results from the previous problem and insert (5.54) in (5.57).

$$(5.57) = \frac{1}{\ell_t(\beta)} \left( \left[ \begin{array}{c} p_t(1-p_t) \\ p_t(1-p_t)z_t \end{array} \right] [(y_t p_t^{y_t-1} (1-p_t)^{1-y_t} - p_t^{y_t} (1-y_t)(1-p_t)^{-y_t})] \right) \quad (5.58)$$

$$= \frac{1}{p_t^{y_t} (1-p_t)^{1-y_t}} \left[ \begin{array}{c} y_t p_t^{y_t} (1-p_t)^{2-y_t} - p_t^{y_t+1} (1-y_t)(1-p_t)^{1-y_t} \\ y_t p_t^{y_t} (1-p_t)^{2-y_t} z_t - p_t^{y_t+1} (1-y_t)(1-p_t)^{1-y_t} z_t \end{array} \right] \quad (5.59)$$

$$= \left[ \begin{array}{c} \frac{p_t^{y_t} (1-p_t)^{1-y_t}}{p_t^{y_t} (1-p_t)^{1-y_t}} \left[ \begin{array}{c} y_t(1-p_t) - p_t(1-y_t) \\ y_t(1-p_t)z_t - p_t(1-y_t)z_t \end{array} \right] \end{array} \right] \quad (5.60)$$

$$= \left[ \begin{array}{c} y_t - y_t p_t - p_t + y_t p_t \\ z_t(y_t - y_t p_t - p_t + y_t p_t) \end{array} \right] \quad (5.61)$$

$$= \left[ \begin{array}{c} y_t - p_t \\ z_t(y_t - p_t) \end{array} \right] \quad (5.62)$$

$$= (y_t - p_t)x_t \quad (5.63)$$

This concludes the derivation of the score contribution for observation  $t \in \{1, 2, \dots, T\}$  with respect to  $\beta$ .

**(3)** Note that since  $y_t|x_t$  follows a Bernoulli distribution we have that the expected value is equal to the probability that  $y_t$  takes on the value 1:  $\mathbb{E}[y_t|x_t] = \text{prob}(y_t = 1|x_t) = p_0$ . Along with the results from the previous problem we have the first order conditions in terms of the true parameters.

$$\mathbb{E} \left[ \left. \frac{\partial \log \ell_t(\beta)}{\partial a} \right|_{\beta=\beta_0} \right] = \mathbb{E} \left[ y_t - \frac{1}{1 + \exp(-a_0)} \right] \quad (5.64)$$

$$= \mathbb{E}[y_t - p_0] = \mathbb{E}[\mathbb{E}[y_t - p_0|x_t]] = \mathbb{E}[p_0 - p_0] = 0 \quad (5.65)$$

$$\mathbb{E} \left[ \left. \frac{\partial \log \ell_t(\beta)}{\partial b} \right|_{\beta=\beta_0} \right] = \mathbb{E} \left[ z_t \left( y_t - \frac{1}{1 + \exp(-a_0)} \right) \right] = \mathbb{E}[y_t z_t] - \mathbb{E} \left[ \frac{z_t}{1 + \exp(-a_0)} \right] \quad (5.66)$$

$$= \mathbb{E}[z_t(y_t - p_0)] = \mathbb{E}[\mathbb{E}[z_t(y_t - p_0)|x_t]] = \mathbb{E}[z_t \mathbb{E}[p_0 - p_0]] = 0 \quad (5.67)$$

We can interpret (5.65) as the expected residual (deviation from the conditinal expectation). This is clearly seen by a simple decomposing of the stochastic variable  $y_t$ :

$$y_t = \mathbb{E}[y_t|x_t] + \epsilon_t \iff \mathbb{E}[\epsilon_t|x_t] = \underbrace{\mathbb{E}[y_t - \mathbb{E}[y_t|x_t]|x_t]}_{\epsilon_t} = 0 \quad ^{15} \quad (5.68)$$

In the context of the problem  $\epsilon_t = y_t - p_0$  which is the error term or residual of  $y_t$ .

The interpretation of (5.67) is that there is no correlation between  $z_t$  and the residuals. This is seen directly by writing

$$\mathbb{E}[z_t(y_t - p_0)] = \mathbb{E}[z_t\epsilon_t] - \mathbb{E}[z_t]\mathbb{E}[\epsilon_t] = \mathbb{E}[z_t\epsilon_t] - \mathbb{E}[z_t] \cdot 0 = Cov(y_t, \epsilon_t) \quad (5.69)$$

The first order conditions in terms of  $\{y_t, x_t\}_{t=1}^T$  and the population parameters are given by

$$\mathbb{E} \left[ \left. \frac{\partial \log L_t(\beta)}{\partial \beta} \right|_{\beta=\beta_0} \right] = \mathbb{E} \left[ \sum_{t=1}^T \left. \frac{\partial \log \ell_t(\beta)}{\partial \beta} \right|_{\beta=\beta_0} \right] = \mathbb{E} \left[ \sum_{t=1}^T s_t(\beta_0) \right] = 0 \quad (5.70)$$

**Comparing with OLS** Ordinary least Squares has the well known moment conditions  $\mathbb{E}[x_t(y_t - x_t'\alpha)] = 0$ . The interpretation of these conditions are the same as for the model considered currently: The explanatory variables are uncorrelated with residuals. However  $x_t'\alpha$  is a linear function, which  $p_t$  is not and therefore we cannot estimate  $\beta$  using OLS.

**Convergence of score to expected value.** For (5.71) to hold we need a LLN to apply to the score contributions.

$$\frac{1}{T} \sum_{t=1}^T s_t(\beta_0) \xrightarrow{p} \mathbb{E}[s_t(\beta_0)], \quad T \rightarrow \infty \quad (5.71)$$

For  $\beta = \beta_0$   $y_t$  is identically and independently distributed with finite moments, however for  $z_t$  we do not know if its identically distributed for  $t = 1, 2, \dots, T$ . Therefore it is not trivial to apply the well know LLN result for i.i.d processes - unless we assume that  $z_t$  is identically distributed  $t = 1, 2, \dots, T$  and then (5.71) holds.

From here we derive the conditions for the score contributions to be stationary. Recall that a LLN applies to a weakly stationary (or covariance stationary) and weakly dependent process. The latter holds since the assignment states that  $(y_t, z_t)'$  are independent across observations,  $t = 1, 2, \dots, T$ . Therefore we want to show that  $s_{t,1} = y_t - p_0$  and  $s_{t,2} = z_t(y_t - p_0)$  are weakly stationary.

By direct calculation we obtain the following

$$\mathbb{E}[s_{t,1}] = \mathbb{E}[y_t - p_0] = \mathbb{E}[y_t] - p_0 = p_0 - p_0 = 0 \quad (5.72)$$

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<sup>15</sup>Sometimes called *the CEF-decomposition property* like in theorem 3.1.1. in the famous book by Angrist and Pischke (2009): Mostly Harmless Econometrics: An Empiricist's Companion

Since  $y_t \sim \text{Bernoulli}(p_0)$  we have the well known variance given by

$$\text{Var}(s_{t,1}) = \mathbb{E}[(y_t - \mathbb{E}[y_t])^2] = p_0(1 - p_0) \quad (5.73)$$

and finally because  $y_t$  and  $y_{t-h}$  are independent we have

$$\text{Cov}(s_{t,1}, s_{t-h,1}) = \text{Cov}(y_t - p_0, y_{t-h} - p_0) = 0 \quad (5.74)$$

From (5.72), (5.73) and (5.74) we conclude that  $s_{t,1} = y_t - p_0$  is weakly stationary (or simply that  $y_t$  is).

To show that  $s_{t,2}$  is weakly stationary we first demonstrate that  $y_t$  and  $z_t$  are independent for  $\beta = \beta_0$ . The density function evaluated at  $\beta_0$  is given by

$$f(y_t|x_t; \beta_0) = p_0^{y_t}(1 - p_0)^{1-y_t} \quad (5.75)$$

$$= \left( \frac{1}{1 + \exp(-a_0)} \right)^{y_t} \left( 1 - \left( \frac{1}{1 + \exp(-a_0)} \right) \right)^{1-y_t} \quad (5.76)$$

$$= f(y_t|a_0) \quad (5.77)$$

, for all values of  $z_t$ . Hence  $y_t$  and  $z_t$  are independent for  $\beta = \beta_0 = (a_0, b_0)' = (a_0, 0)'$ .

In the following we utilize (5.72) to calculate the relevant moments of  $s_{t,2}$ :

$$\mathbb{E}[s_{t,2}] = \mathbb{E}[z_t(y_t - p_0)] \stackrel{LIE}{=} \mathbb{E}[\mathbb{E}[z_t(y_t - p_0)|z_t]] = \mathbb{E}[z_t \mathbb{E}[y_t - p_0|z_t]] = \mathbb{E}[z_t \cdot 0] = 0 \quad (5.78)$$

$$\text{Var}(s_{t,2}) = \text{Var}(z_t(y_t - p_0)) = \mathbb{E}[z_t^2(y_t - p_0)^2] - \mathbb{E}[z_t] \mathbb{E}[(y_t - p_0)] \quad (5.79)$$

$$= \mathbb{E}[\mathbb{E}[z_t^2(y_t - p_0)^2|z_t]] \quad (5.80)$$

$$= \mathbb{E}[z_t^2 \mathbb{E}[(y_t - p_0)^2|z_t]] \quad (5.81)$$

$$\stackrel{indep.}{=} \mathbb{E}[z_t^2] \mathbb{E}[\mathbb{E}[(y_t - p_0)^2|z_t]] \quad (5.82)$$

$$= \mathbb{E}[z_t^2] \mathbb{E}[(y_t - p_0)^2] \quad (5.83)$$

$$= \mathbb{E}[z_t^2] p_0(1 - p_0) \quad (5.84)$$

, which is well defined since we have  $\mathbb{E}[z_t^2] < \infty$ . For the same reasons as in (5.74) we have

$$\text{Cov}(s_{t,2}, s_{t-h,2}) = \text{Cov}(z_t(y_t - p_0), z_{t-h}(y_{t-h} - p_0)) = 0 \quad (5.85)$$

From (5.78), (5.84) and (5.85) we conclude that  $s_{t,2} = z_t(y_t - p_0)$  is weakly stationary if  $\mathbb{E}[z_t^2] = \text{constant}$ . In this case we have that a LLN applies to  $s_t(\beta_0)$  and (5.71) holds.

(4) Recall from (5.63) that the score contribution is

$$s_t(\beta) = \begin{bmatrix} y_t - p_t \\ z_t(y_t - p_t) \end{bmatrix} = (y_t - p_t)x_t \quad (5.86)$$

The Hessian contribution is given by the derivative of the score with respect to  $\beta'$

$$H_t(\beta) = \frac{\partial s_t(\beta)}{\partial \beta'} = \frac{\partial}{\partial \beta'}(y_t - p_t)x_t \quad (5.87)$$

Since  $y_t$  does not depend on  $\beta'$ , the Hessian contribution simplifies to:

$$H_t(\beta) = -\frac{\partial}{\partial \beta'}(p_t)x_t \quad (5.88)$$

we let  $\eta_t(\beta) = x'_t\beta$ , and the chain rule gives us

$$\frac{\partial p_t}{\partial \beta'} = \frac{\partial p_t(\eta_t)}{\partial \eta_t} \cdot \frac{\partial \eta_t(\beta)}{\partial \beta'} \quad (5.89)$$

We start by calculating  $\frac{\partial p_t(\eta_t)}{\partial \eta_t}$

$$\frac{\partial p_t(\eta_t)}{\partial \eta_t} = \frac{\partial}{\partial \eta_t}(1 + \exp(-\eta_t))^{-1} = -(1 + \exp(-\eta_t))^{-2}(-\exp(-\eta_t)) = \frac{\exp(-\eta_t)}{(1 + \exp(-\eta_t))^2} \quad (5.90)$$

Using  $1 - p_t(\eta_t) = \frac{\exp(-\eta_t)}{1 + \exp(-\eta_t)}$  we can rewrite (5.90) into a simpler form

$$\frac{\partial p_t(\eta_t)}{\partial \eta_t} = \frac{\exp(-\eta_t)}{(1 + \exp(-\eta_t))^2} = \frac{1}{1 + \exp(-\eta_t)} \frac{\exp(-\eta_t)}{1 + \exp(-\eta_t)} = p_t(\eta_t)(1 - p_t(\eta_t)) \quad (5.91)$$

Now we look at the second part  $\frac{\partial \eta_t(\beta)}{\partial \beta'}$  where  $\eta_t = x'_t\beta$

$$\frac{\partial \eta_t(\beta)}{\partial \beta'} = \frac{\partial}{\partial \beta'} x'_t\beta = x'_t \quad (5.92)$$

Combining (5.88) and (5.89) we get:

$$H_t(\beta) = -\frac{\partial p_t}{\partial \beta'}x_t = -\frac{\partial p_t(\eta_t)}{\partial \eta_t} \cdot \frac{\partial \eta_t(\beta)}{\partial \beta'}x_t \quad (5.93)$$

Inserting (5.91) and (5.92) into the equation for the hessian contribution.

$$H_t(\beta) = -p_t(1 - p_t) \cdot x'_t \cdot x_t = -p_t(1 - p_t)x'_tx_t \quad (5.94)$$

with

$$x_t' x_t = \begin{bmatrix} 1 & z_t \\ z_t & z_t^2 \end{bmatrix} \quad (5.95)$$

Meaning the total hessian contribution can be defined as:

$$H_t(\beta) = \begin{bmatrix} \frac{\partial^2 \log \ell_t(\beta)}{\partial a \partial a} & \frac{\partial^2 \log \ell_t(\beta)}{\partial a \partial b} \\ \frac{\partial^2 \log \ell_t(\beta)}{\partial b \partial a} & \frac{\partial^2 \log \ell_t(\beta)}{\partial b \partial b} \end{bmatrix} = -p_t(1-p_t) \begin{bmatrix} 1 & z_t \\ z_t & z_t^2 \end{bmatrix} \quad (5.96)$$

$$= \begin{bmatrix} -p_t(1-p_t) & -p_t(1-p_t)z_t \\ -p_t(1-p_t)z_t & -p_t(1-p_t)z_t^2 \end{bmatrix} \quad (5.97)$$

**The information:** By inserting the Hessian contribution into  $I(\beta_0) = -E[H_t(\beta_0)]$ , with  $\beta = \beta_0$  we get the information:

$$I(\beta_0) = -E[H_t(\beta_0)] = E \begin{bmatrix} \frac{\exp(-a_0)}{(1+\exp(-a_0))^2} & \frac{\exp(-a_0)}{(1+\exp(-a_0))^2} z_t \\ \frac{\exp(-a_0)}{(1+\exp(-a_0))^2} z_t & \frac{\exp(-a_0)}{(1+\exp(-a_0))^2} z_t^2 \end{bmatrix} \quad (5.98)$$

We notice that  $\frac{\exp(-a_0)}{(1+\exp(-a_0))^2}$  is a constant not affected by  $z_t$ . This means that it can be drawn out from the expectation

$$I(\beta_0) = -E[H_t(\beta_0)] = \frac{\exp(-a_0)}{(1+\exp(-a_0))^2} \underbrace{\begin{bmatrix} 1 & E(z_t) \\ E(z_t) & E(z_t^2) \end{bmatrix}}_{Z_t} \quad (5.99)$$

**Conditions under which the information is positive definite:** Since the constant  $\frac{\exp(-a_0)}{(1+\exp(-a_0))^2} > 0 \quad \forall a_0 \in \mathbb{R}$  the information is positive definite if the matrix  $Z_t$  is positive definite.

We notice that the matrix  $Z_t$  is a symmetrical 2x2 matrix, which means that it is positive definite if the leading sub-determinants (also called leading principle minors) are positive definite.<sup>16</sup> The first leading sub-determinant:

$$\det([1]) = 1 \quad (5.100)$$

The second leading sub-determinant

$$\det \left( \begin{bmatrix} 1 & E(z_t) \\ E(z_t) & E(z_t^2) \end{bmatrix} \right) = 1 \cdot E(z_t^2) - E(z_t)E(z_t) = E(z_t^2) - E(z_t)^2 \quad (5.101)$$

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<sup>16</sup>"Forelæsningsnoter til Lineær Algebra", Niels Vigand Pedersen, theorem 7.5.5

This is the variance formula

$$E(z_t^2) - E(z_t)^2 = \text{Var}(z_t) \quad (5.102)$$

This means that if  $\text{Var}(z_t) > 0$  then it must hold that the information is positive definite. This means that information about  $\beta_0$  only depends on the variation of  $z_t$ . If there is any variation in the variable  $z_t$ , then we know something about  $\beta_0$ .

**What changes if  $b_0 \neq 0$**  The information now returns to the original form, where  $p_t(1 - p_t) = \frac{\exp(-(a_0 + b_0 z_t))}{(1 + \exp(-(a_0 + b_0 z_t)))^2}$  now is dependent on  $z_t$ :

$$I(\beta_0) = -E[H_t(\beta_0)] = E[p_t(1 - p_t)Z_t] \quad (5.103)$$

It still holds that there needs to be variation in  $z_t$  to make the information positive definite. But there is now an additional factor dependent on  $z_t$ . The only thing that could make the information non positive definite would be if  $p_t(1 - p_t) = 0$ . If  $p_t \approx 0$  or  $p_t \approx 1$  then  $p_t(1 - p_t) \approx 0$ . This would make  $I(\beta_0) \approx 0$  which means that the model provides essentially no information about  $\beta_0$ .

(5) As stated in Theorem 3.1<sup>17</sup>, the (Q)MLE is asymptotically normally distributed provided that Assumption 3.1<sup>18</sup> holds.

$$\sqrt{T}(\hat{\beta} - \beta_0) \xrightarrow{d} N[0, \Omega_\beta] \Rightarrow \begin{cases} \sqrt{T}(\hat{a} - a_0) \xrightarrow{d} N[0, \Omega_{\beta_{11}}] \\ \sqrt{T}(\hat{b} - b_0) \xrightarrow{d} N[0, \Omega_{\beta_{22}}] \end{cases} \quad (5.104)$$

where  $\Omega_\beta = \mathcal{I}(\beta_0)^{-1}$  under the assumption that the model is correctly specified and that the distributional assumption holds (as assumed in the description of the assignment).  $\Omega_{\beta_{11}}$  and  $\Omega_{\beta_{22}}$  are diagonal elements in the covariance matrix. First, we use the information found in the previous question to calculate the inverse information (covariance matrix). Recall that  $(cA)^{-1} = \frac{1}{c}A^{-1}$  if  $c$  is a scalar and  $A^{-1} = \frac{1}{ab-cd} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ .

$$\Omega_\beta = \mathcal{I}(\beta_0)^{-1} = \left[ \frac{\exp(-a_0)}{[1 + \exp(-a_0)]^2} \right]^{-1} \cdot \frac{1}{E(z_t^2) - E[z_t]E[z_t]} \begin{pmatrix} E[z_t^2] & -E[z_t] \\ -E[z_t] & 1 \end{pmatrix} \quad (5.105)$$

$$= \frac{[1 + \exp(-a_0)]^2}{\exp(-a_0)} \cdot \frac{1}{E(z_t^2) - E[z_t]E[z_t]} \begin{pmatrix} E[z_t^2] & -E[z_t] \\ -E[z_t] & 1 \end{pmatrix} \quad (5.106)$$

$$= \frac{1}{p_0(1 - p_0)\text{Var}(z_t)} \begin{pmatrix} E[z_t^2] & -E[z_t] \\ -E[z_t] & 1 \end{pmatrix} \quad (5.107)$$

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<sup>17</sup>Nielsen (2025)

<sup>18</sup>Nielsen (2025)

Now, I use the asymptotic variance to derive the distribution of  $\sqrt{T}\hat{b}$ .

$$\sqrt{T}(\hat{b} - b_0) \xrightarrow{d} N \left[ 0, \frac{1}{p_0(1-p_0)\text{Var}(z_t)} \right] \quad (5.108)$$

Recall that  $b_0 = 0$  which implies that  $\sqrt{T}(\hat{b} - b_0) = \sqrt{T}(\hat{b} - 0) = \sqrt{T}\hat{b}$ . Therefore  $\sqrt{T}\hat{b}$  is asymptotically normally distributed as follows.

$$\sqrt{T}\hat{b} \xrightarrow{d} N \left[ 0, \frac{1}{p_0(1-p_0)\text{Var}(z_t)} \right] \quad (5.109)$$

This implies that the variance of the estimator decreases as the variance of the regressor increases.

**(6)** The natural extension would be to assume that  $y_t$  follows a conditional Bernoulli distribution conditional on  $w_t = (1, z_t, z_{t-1}, p_{t-1})'$ . The proposed model is specified by (5.110) and (5.111).

$$y_t | x_t, x_{t-1}, p_{t-1} \stackrel{d}{=} \text{Bernoulli}(p_t), \quad t = 1, 2, \dots, T \quad (5.110)$$

with

$$p_t = \frac{1}{1 + \exp(-w_t' \delta)} \quad (5.111)$$

where  $\delta = (\delta_0, \delta_1, \delta_2, \delta_3)'$  and  $x_0, p_0$  are given.

In short we now model the stochastic variable  $y_t \in \{0, 1\}$  as a conditional Bernoulli process, where the conditional probability of success conditional on  $w_t = (x_t, x_{t-1}, p_{t-1})'$ :  $\text{prob}(y_t = 1 | x_t, x_{t-1}, p_{t-1})$ , is given by the logistic function in (5.111).

To obtain a consistent estimator of the parameters  $\delta$  using (Q)MLE the likelihood function must take the time dependency into account. We simply modify the likelihood function to condition on  $w_t$ , by including the new form for  $p_t$ . To be explicit we write out the log-likelihood function:

$$\log L_T(\delta) = \sum_{t=1}^T \log \ell_t(\delta) \quad (5.112)$$

$$= \sum_{t=1}^T \left[ y_t \log(p_t) + (1 - y_t) \log(1 - p_t) \right] \quad (5.113)$$

$$= \sum_{t=1}^T \left[ y_t \log \left( \frac{1}{1 + \exp(-w'_t \delta)} \right) + (1 - y_t) \log \left( \frac{1 + \exp(-w'_t \delta) - 1}{1 + \exp(-w'_t \delta)} \right) \right] \quad (5.114)$$

$$= \sum_{t=1}^T \left[ -y_t \log(1 + \exp(-(w'_t \delta))) - (1 - y_t)(w'_t \delta + \log(1 + \exp(-w'_t \delta))) \right] \quad (5.115)$$

Now we have demonstrated how the model could be modified to accommodate the time series observations.

### 5.3 Literature

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